

SHUAI ZHOU

Master Student of Robotics

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EDUCATION

CARNEGIE MELLON UNIVERSITY

Master of Science in Robotics

Pittsburgh, USA

Aug 2026 — Jun 2028

SOUTH CHINA UNIVERSITY OF TECHNOLOGY

Bachelor of Engineering in Robotics; GPA: 3.86/4.00

Guangzhou, China

Aug 2022 — Jun 2026

UNIVERSITY OF CALIFORNIA, BERKELEY

Exchange Student in EECS; GPA: 4.00/4.00

Berkeley, USA

Aug 2023 — Dec 2023

EXPERIENCE

HKUST (Guangzhou), IPRN Lab

Research Topics: Foundation Model, Active Perception, Egocentric Human Data.

Guangzhou, China

Mar 2026 — Aug 2026

Advisor: *Haoang Li*

- Active Perception with AVP-ALOHA teleoperation system, UMI and Egocentric-Human video pretraining.
- [Egocentric-Human video subtask annotation pipeline](#) and scaling foundation model with multi-source data.

CARNEGIE MELLON UNIVERSITY, SAFE AI Lab

Research Topics: Manipulation, Egocentric Human Data, Whole-Body Controller.

Pittsburgh, USA

Aug 2025 — Mar 2026

Advisor: *Ding Zhao*

- Learning versatile humanoid manipulation policy and developing robust lower-body controller [4].
- Egocentric Cross-Embodiment manipulation with tactile latent prediction [2].

CARNEGIE MELLON UNIVERSITY, ARCS Lab

Research Topics: Learning for Planning, Multi-Robot System.

Pittsburgh, USA

Apr 2025 — Mar 2026

Advisor: *Jiaoyang Li*

- Collaborative multi-robot non-prehensile manipulation via flow-matching model and lazy planning [3].
- Learning execution-time prediction models for closed-loop multi-robot coordination and dependency optimization [1].

SHANGHAI JIAO TONG UNIVERSITY, RAP Lab

Research Topics: Heuristic Search, Multi-Robot System.

Shanghai, China

Apr 2024 — Apr 2025

Advisor: *Zhongqiang Ren*

- Scalable planners for Multi-Robot Path Planning that scale to 1,000+ robots [6].
- Complete, anytime planning framework with pruning and sorting strategies that accelerated search [5].

PUBLICATION

PREPRINT

- [1] **From Discrete Plans to Real-World Execution: A World-Model-Driven Framework for Execution-Aware Multi-Agent Path Finding**

Jingtian Yan, Shuai Zhou, He Jiang, Stephen Smith, Jiaoyang Li. **Under Review, 2026** [\[Paper\]](#)

- [2] **Egocentric Cross-Embodiment Manipulation with Embodiment Dreaming**

Binghong Chen*, Yaru Niu*, Changwei Yao*, Zhenlong Fang*, Shangtao Li*, Revanth Krishna Senthilkumaran, [Shuai Zhou](#), Yuemin Mao, Hao Zhang, Bingqing Chen, Chen Qiu, Hongtei (Eric) Tseng, Changliu Liu, Jonathan Francis, Ding Zhao. **Under Review, 2026** [\[Paper\]](#)

PUBLISHED

- [3] **LAMP: Long-Horizon Adaptive Manipulation Planning for Multi-Robot Collaboration in Cluttered Space**
[Shuai Zhou](#), Yorai Shaoul, Jiaoyang Li. **IROS, 2026**. [\[Paper\]](#) [\[Project\]](#)

- [4] **HTD: Learning Versatile Humanoid Manipulation with Touch Dreaming**

Yaru Niu, Zhenlong Fang, Binghong Chen, [Shuai Zhou](#), Revanth Krishna Senthilkumaran, Hao Zhang, Bingqing Chen, Chen Qiu, Hongtei (Eric) Tseng, Jonathan Francis, Ding Zhao. **IROS, 2026**. [\[Paper\]](#) [\[Project\]](#)

- [5] **LSRP*: Scalable and Anytime Planning for Multi-Agent Path Finding with Asynchronous Actions**

[Shuai Zhou](#), Shizhe Zhao, Zhongqiang Ren. **SoCS, 2025**. [\[Paper\]](#) [\[Project\]](#)

- [6] **Loosely Synchronized Rule-Based Planning for Multi-Agent Path Finding with Asynchronous Actions**

[Shuai Zhou](#), Shizhe Zhao, Zhongqiang Ren. **AAAI, 2025**. [\[Paper\]](#) [\[Project\]](#)

SKILLS

Platforms: AgileX Aloha, AgileX Piper Arm, Apple Vision Pro, UMI, Unitree G1 Humanoid

Simulators: IsaacLab, MuJoCo, PyBullet

Programming: Python, C++, Java, HTML, MATLAB, $\text{\LaTeX}$

SERVICES

Conference Reviewer: 2025-2026 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)

Journal Reviewer: 2026 IEEE Robotics and Automation Letters (RA-L)

HONORS

Merit Student of South China University of Technology

Top student in Robotics Engineering, Class of 2022

Guangzhou, China

Feb 2024